Primate Evolution



Intro

What are Primates?

Classifying Primates

Characteristics of Primates

What are Primates?

- The Order of animals that includes Humans
- Also includes:
  - Great Apes
  - Gibbons
  - Monkeys
  - Lemurs
  - Lorises
  - Tarsiers
- Non-human primates are our closest living relatives, so can compare to see how human characteristics may have evolved.
  - Comparative anatomy
  - Comparative biochemistry
  - Behaviour of living primates
  - Fossils of primates

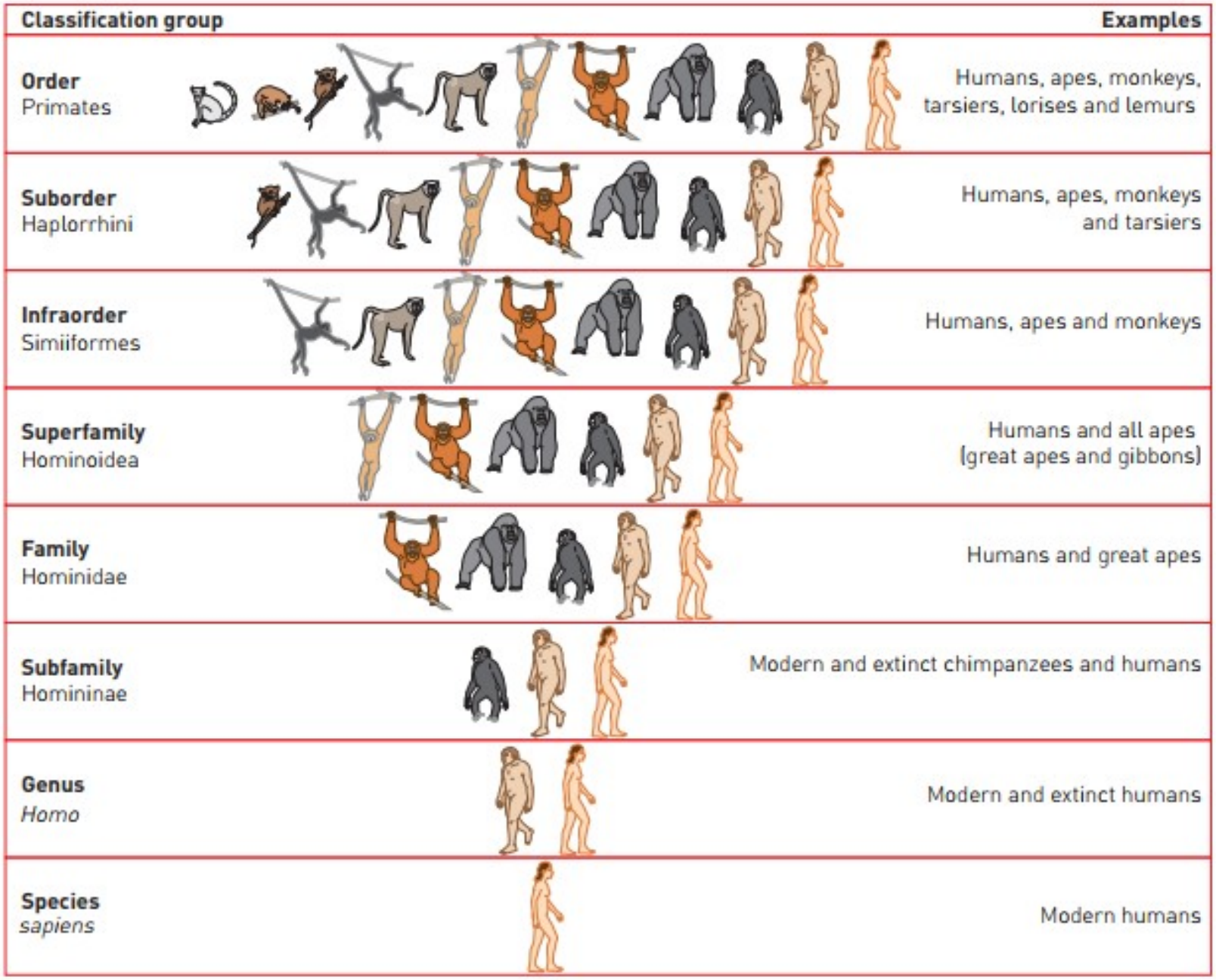


Figure 17.2 A diagrammatic representation of the hierarchy within the primate order

Classification and hierarchy

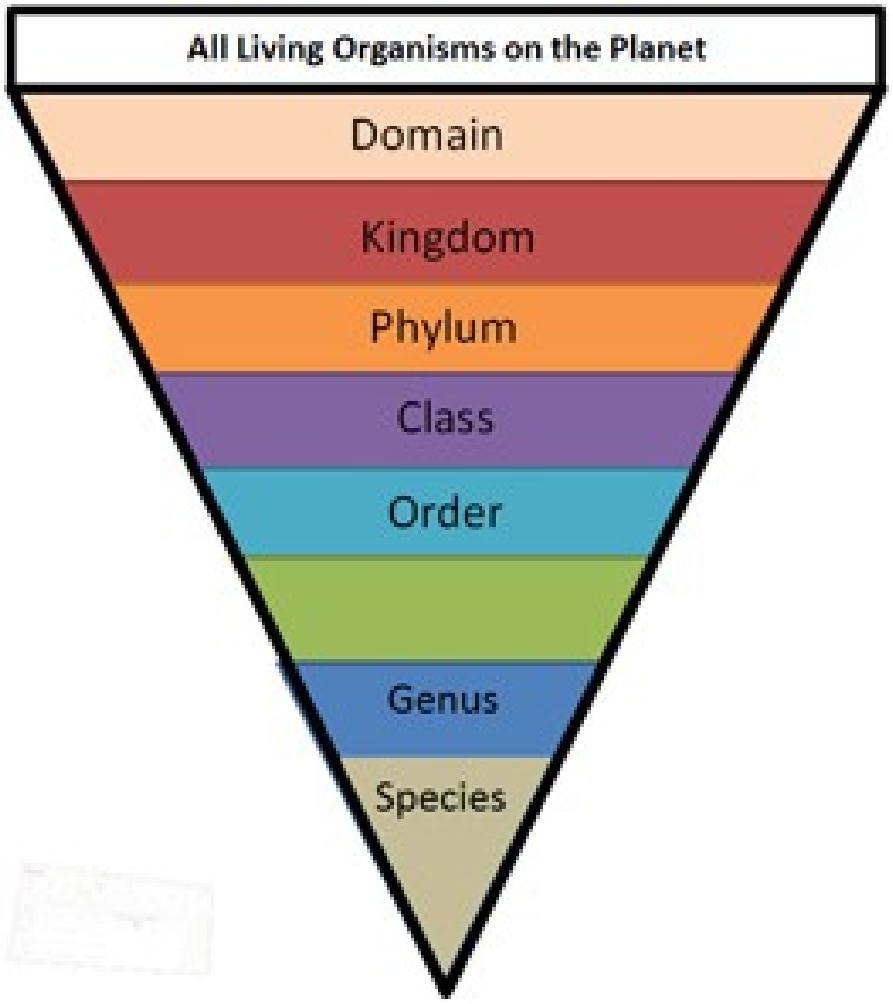


Table 17.1 The classification of humans within the primate order

Level of classification	Name	Examples
Order	Primates	Primates include tarsiers, lemurs, lorises, monkeys, apes and humans
Suborder	Haplorrhini	Haplorrhini include tarsiers, monkeys, apes and humans
Infraorder	Simiiformes	Simiiformes include monkeys, apes and humans
Parvorder	Catarrhini	Catarrhines include Old World monkeys, apes and humans
Superfamily	Hominoidea	Hominoids include apes and humans
Family	Hominidae	Hominids include all modern and extinct orang-utans, gorillas, chimpanzees and humans
Subfamily	Homininae	Hominines include all modern and extinct chimpanzees and humans
Tribe*	Hominini	Hominins include extinct ancestors of humans and modern humans
Genus	*Homo*	*Homo* includes some extinct ancestors of humans and modern humans
Species	*sapiens*	*Homo sapiens* are modern humans

**Note*: Tribe is a classification group within a subfamily. The meaning of tribe here is different from the use of tribe to describe an ethnic group of people.

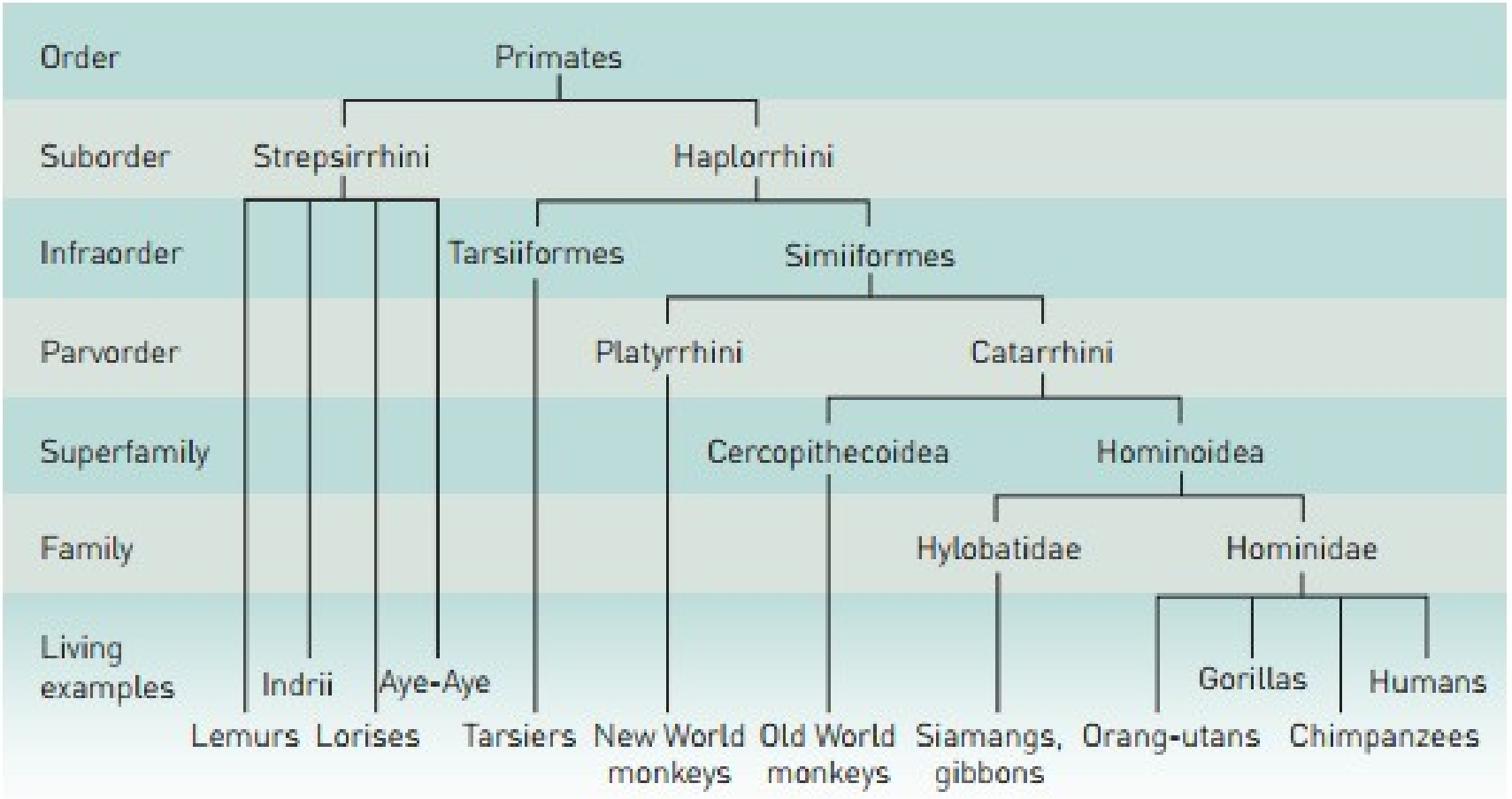


Figure 17.3 A simplified classification of the primates

Classifying primates

- Classification is a dynamic process: as new information comes to light, classification of an organism may change.
- Eg:
  - Earlier classification divided primates into two groups:
    - Simians (humans and human-like primates)
    - Prosimians (primitive primates)
    - Tarsiers were included with prosimians, based on physical characteristics
  - Based on studies of protein sequences and DNA:
    - Most scientists now classify Tarsiers as more closely related to monkeys, apes and humans
  - Humans: classified in same family as great apes – we have very similar DNA and many shared characteristics.

Primate Characteristics

- Main features: combination of grasping fingers and toes, and overlapping binocular vision - distinctive to primates.

Table 17.2 A summary of the characteristics of members of the order Primates

Feature	Primate characteristics
Body	Not specialised for a particular environment
Limbs	Generally unspecialised
Hands/feet	Pentadactyl – five fingers or toes Nails instead of claws Grasping fingers and toes with friction ridges for gripping First digit opposable
Eyes	Forward facing for three-dimensional (stereoscopic) vision Most are able to distinguish colour
Sense of smell	Very poor
Teeth	Four incisors in both the upper and lower jaw
Brain	Large and complex Cerebrum size increases as primates become more highly evolved
Reproduction	Not restricted to a breeding season Rhythmical sexual cycle Usually only one offspring at a time Long period of parental care for offspring